

A Complex Continuous Normal Convex-Cyclic Operator

Literature-implied answer to the continuous-normal half of Question 2 in arXiv:1410.4664

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17 August 2026

Status. *Literature-implied answer (complex continuous-normal subproblem).* The argument below gives an explicit affirmative example on a complex Hilbert space. Its only external mathematical input is the real convex Stone–Weierstrass theorem of Feldman and McGuire. The pure-hyponormal alternative in the source question is not settled here.

1 Source question

Bermúdez, Bonilla, and Feldman ask in Section 7, Question 2 of [1]:

Are there pure hyponormal operators or continuous normal operators that are convex-cyclic?

The question occurs on page 17 of the locally rendered arXiv source.

Sanders [24] proved that if $T : H \rightarrow H$ is a hyponormal operator on a Hilbert space H , then T is not weakly hypercyclic. A hyponormal operator is *pure* if its restriction to any of its reducing subspaces is not normal. That is, a hyponormal operator T is pure if T cannot be written in the form $T = S \oplus N$ where N is a normal operator.

Question 2. Are there pure hyponormal operators or continuous normal operators that are convex-cyclic?

Here a polynomial $p(z) = \sum_{n=0}^N c_n z^n$ is a *convex polynomial* when $c_n \geq 0$ and $\sum c_n = 1$. An operator T is convex-cyclic if some vector x has $\{p(T)x : p \in \mathcal{CP}\}$ dense in the ambient space. A normal operator is called continuous here when it has no eigenvalues (equivalently, its scalar spectral measure has no atoms in the multiplicity-one example below).

2 Literature input and identification

Feldman and McGuire prove that if ρ is a compactly supported finite positive measure on $\mathbb{R}$, then real convex polynomials are dense in $L^2_{\mathbb{R}}(\rho)$ exactly when $\rho([-1, \infty)) = 0$; see [2, Theorem 7.1]. Their theorem is stated for a real Hilbert space and therefore does not, by itself, make multiplication by a negative real variable convex-cyclic on a complex L^2 space. The even–odd construction below supplies the missing complex direction.

3 Idea of the proof

Put the spectrum on the imaginary ray $i[a, b]$ with $a > 1$. Even powers of ir are powers of the negative real variable $-r^2$, so the real theorem approximates the real part of an arbitrary target. Odd powers are ir times the same negative-real powers, so they approximate the imaginary part after division by the bounded nonvanishing function r . Averaging an even convex polynomial and an odd convex polynomial preserves nonnegative coefficients and total coefficient mass one.

4 The result

Theorem 1. *Let $1 < a < b < \infty$, let μ be a finite nonatomic positive Borel measure on $[a, b]$, and let*

$$H = L^2([a, b], \mu; \mathbb{C}), \quad (Tf)(r) = irf(r).$$

Then T is a continuous normal convex-cyclic operator. The constant function $\mathbf{1}$ is a convex-cyclic vector. If $\text{supp } \mu = [a, b]$, then $\sigma(T) = i[a, b]$ and $\sigma_p(T) = \sigma_p(T^) = \emptyset$.*

Proof. Multiplication by the bounded measurable function $r \mapsto ir$ is normal. If $Tf = \lambda f$, then $(ir - \lambda)f(r) = 0$ almost everywhere. The zero set of $ir - \lambda$ contains at most one point, and nonatomicity of μ forces $f = 0$. The same argument applies to T^* , which is multiplication by $-ir$. Thus T has no eigenvalues. The assertion about the spectrum follows from the usual spectrum formula for multiplication operators.

It remains to prove convex-cyclicity. Push μ forward under $r \mapsto x = -r^2$ and call the resulting measure ρ . Its support is contained in

$$[-b^2, -a^2] \subset (-\infty, -1),$$

so [2, Theorem 7.1] says that convex polynomials are dense in the real space $L^2_{\mathbb{R}}(\rho)$.

Fix $F = U + iV \in H$, with U, V real-valued, and fix $\varepsilon > 0$. Since $r \geq a > 0$, both $2U(r)$ and $2V(r)/r$ belong to $L^2_{\mathbb{R}}(\mu)$. By the cited density theorem there are convex polynomials q and s such that

$$\begin{aligned} \|q(-r^2) - 2U(r)\|_{L^2(\mu)} &< \varepsilon, \\ \|s(-r^2) - 2V(r)/r\|_{L^2(\mu)} &< \varepsilon/b. \end{aligned}$$

Define

$$p(z) = \frac{1}{2}q(z^2) + \frac{1}{2}zs(z^2).$$

The coefficients of $q(z^2)$ occupy the even degrees, the coefficients of $zs(z^2)$ occupy the odd degrees, and each coefficient family is nonnegative with sum one. Hence p is a convex polynomial. On the spectral ray,

$$p(ir) = \frac{1}{2}q(-r^2) + \frac{i}{2}rs(-r^2).$$

Real and imaginary parts are orthogonal pointwise, so

$$\begin{aligned} \|p(ir) - F(r)\|_2^2 &= \frac{1}{4}\|q(-r^2) - 2U(r)\|_2^2 + \frac{1}{4}\|rs(-r^2) - 2V(r)\|_2^2 \\ &< \frac{1}{4}\varepsilon^2 + \frac{1}{4}b^2(\varepsilon/b)^2 = \frac{1}{2}\varepsilon^2. \end{aligned}$$

As F and ε were arbitrary, the convex polynomials in ir are dense in H . Since $p(T)\mathbf{1} = p(ir)$, the vector $\mathbf{1}$ is convex-cyclic. $\square$

Corollary 2. *On $L^2([2, 3], dr; \mathbb{C})$, the operator*

$$(Tf)(r) = irf(r)$$

is an explicit complex continuous normal convex-cyclic operator with spectrum $i[2, 3]$ and empty point spectrum for both T and T^ .*

5 Verification and scope

The script `code/check_even_odd_reduction.py` checks, using exact rational arithmetic over 1,000 randomized pairs of convex polynomials, that $\frac{1}{2}(q(z^2) + zs(z^2))$ remains convex and that its evaluation at $z = ir$ has exactly the even/odd form used above. This is a transcription check, not a substitute for the density theorem.

The construction settles the continuous-normal existence alternative on a *complex* Hilbert space and gives a whole nonatomic family. It does not produce a pure hyponormal nonnormal operator, does not address whether every convex-cyclic operator has densely many convex-cyclic vectors, and does not answer how large $\sigma_p(T^*)$ can be (the example has empty adjoint point spectrum).

6 Search and provenance note

A bounded search used the exact wording of Question 2 together with “convex-cyclic”, “continuous normal”, “pure hyponormal”, “point spectrum”, and the authors’ names in the local arXiv corpus and in web/arXiv search. It recovered [1], the later real theorem [2], and special-class convex-cyclicity papers, but no source explicitly stating this complex imaginary-ray construction. Because Theorem 7.1 of [2] is the decisive input and the new step is a direct even/odd complexification, this packet is conservatively classified as a literature-implied answer rather than as an original full solution. The local copies of both papers were rendered from the archived arXiv TeX sources because network PDF retrieval was unavailable during packaging.

7 Human-review recommendation

The result is likely valid. Review should focus on the real-to-complex identification: the pushforward measure, division by r , and the verification that the average of the even and odd substitutions is still a convex polynomial. The literature/provenance classification should also be checked against any later paper not returned by the bounded search.

References

- [1] T. Bermúdez, A. Bonilla, and N. Feldman, *On convex-cyclic operators*, arXiv:1410.4664 (2014), Section 7, Question 2.
- [2] N. S. Feldman and P. J. McGuire, *A Convex Stone–Weierstrass Theorem & Applications*, arXiv:1510.08878 (2015), Theorem 7.1.